Design proposal: Mach-Zehnder Interferometer

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I. Introduction

THIS report outlines my proposal for the layout submission as part of the Phot1x design course. The layout will include three asymmetric Mach-zehnder interferometers with delays 40 µm, 80 µm and 120 µm as well as a symmetric Mach Zehnder for comparison. Throughout the design, light will be guided by stip waveguides of size 220 nm x 500 nm. Light will be coupled into the fundamental TE mode of the waveguide via the grating couplers provided as part of the course PDK. Each of the Mach Zehnder interferometers will be based on two of the broadband directional couplers provided as part of the course PDK, so that both inputs and outputs of the interferometer can be addressed. The remaining allocated space will be occupied by four ring resonators, yet to be designed.

II. Theory

A Mach-Zehnder interferometer is typically built out of two elements, one optical splitter and one combiner. The splitter divides the light into two different paths, each travelling towards one of the inputs of the combiner. For the first and second path, we will respectively define the path lengths as L_1 and L_2, the propagation constants as β_1 and β_2 and the attenuation constants as α_1 and α_2. Once they reach the combiner, the fields in the two paths interfere. Assuming E_i to be the field amplitude input into the splitter, the field in one of the outputs of the combiners will be

$$E_{o1} = \frac{E_i}{2}(e^{-(i\beta_1+\frac{\alpha_1}{2})L_1} + e^{-(i\beta_2+\frac{\alpha_1}{2})L_2}) \tag{1}$$

and the corresponding optical intensity will be

$$I_{o1} = \frac{I_i}{4}|e^{-(i\beta_1+\frac{\alpha_1}{2})L_1} + e^{-(i\beta_2+\frac{\alpha_1}{2})L_2}|^2 \tag{2}$$

For the interferometers considered in this proposal, both paths have the same waveguide geometry and they only differ in length. This means the propagation constant is the same in both paths. If we also neglect optical loss in the waveguide, we obtain

$$I_{o1} = \frac{I_i}{2}[1 + \cos(\beta\Delta L)] \tag{3}$$

where $\Delta L = L_1 - L_2$. Since the structures used as combiners are 2x2 directional couplers, both optical outputs are addressable. The intensity in the second output of the interferometer can be found as follows

$$I_{o2} = I_i - I_{o1} = \frac{I_i}{2}[1 - \cos(\beta\Delta L)]. \tag{4}$$

It is worth noting that the propagation constant depends on the wavelength as follows:

$$\beta(\lambda) = \frac{2\pi n(\lambda)}{\lambda} \tag{5}$$

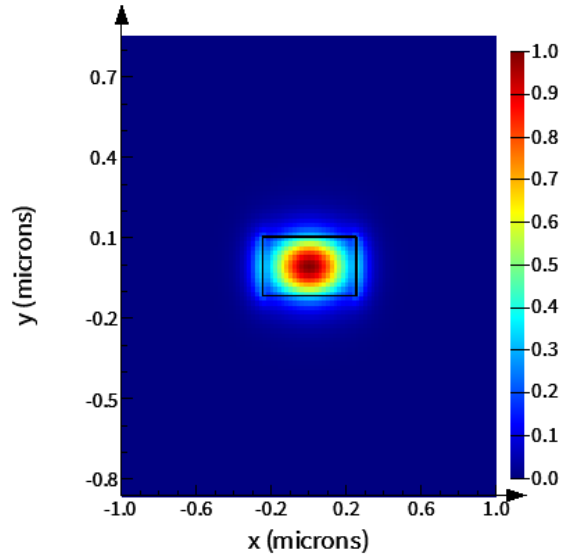

Fig. 1. Simulated mode profile for the fundamental TE mode of the waveguide geometry under consideration

which means that I_{o1} will also change as a function of the wavelength, unless $\Delta L = 0$, oscillating periodically between maxima and minima. The period of these oscillations is called a *Free Spectral Range* (FSR).

Let us consider two wavelength λ_m and λ_{m+1} that are exactly one free spectral range apart. The variation in β will be given by

$$\Delta\beta = \beta_m - \beta_{m+1} \approx -\frac{d\beta}{d\lambda}\Delta\lambda \tag{6}$$

where

$$\frac{d\beta}{d\lambda} = -\frac{2\pi}{\lambda^2}(n - \frac{dn}{d\lambda}\lambda) = -\frac{2\pi}{\lambda^2}n_g \tag{7}$$

and n_g is the group index of the waveguide. Note that the condition of being exactly one FSR apart translates into

$$\Delta\beta\Delta L = 2\pi. \tag{8}$$

Substituting $\Delta\beta$ in here, we can find the expression of the FSR as a function of n_g and ΔL:

$$\Delta\lambda = \frac{\lambda^2}{\Delta L n_g} \tag{9}$$

Using this expression, one can measure the group index of a specific waveguide geometry by fabricating interferometers with specific delays.

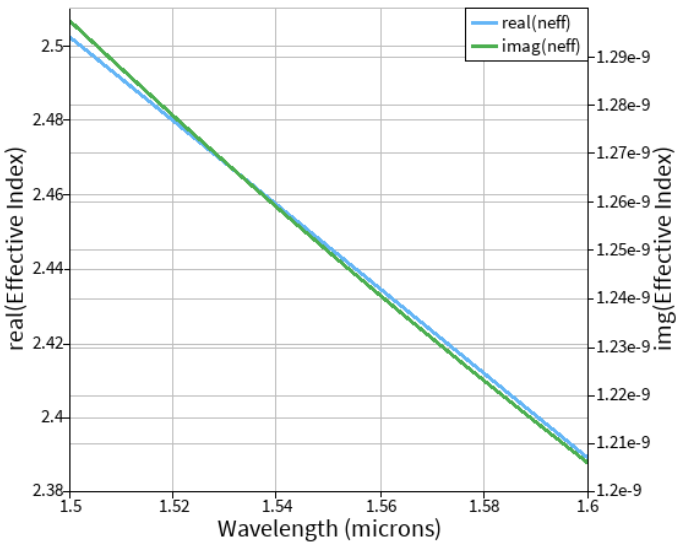

Fig. 2. Simulated values of real and imaginary parts of the effective refractive index as a function of wavelength

III. Modelling and Simulations

In this section we report the simulated properties of the waveguides used in the design, as well as the expected transfer functions of the interferometers.

A. Waveguide properties

As mentioned in the introduction, the waveguides used in this design will be strip waveguides of 220 nm height and 500 nm width, and light will travel in the fundamental TE mode. Figure 1 shows the TE mode profile obtained by simulating this waveguide geometry with Lumerical for a wavelength of 1550 nm. Figure 2 displays the real and imaginary parts of the effective refractive index for the same mode as a function of wavelength and Figure 3 the wavelength dependence of the group index.

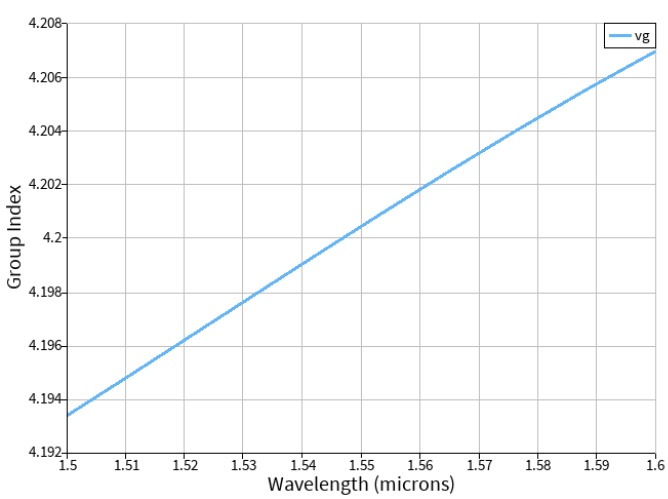

Fig. 3. Simulated values of the group index as a function of wavelength

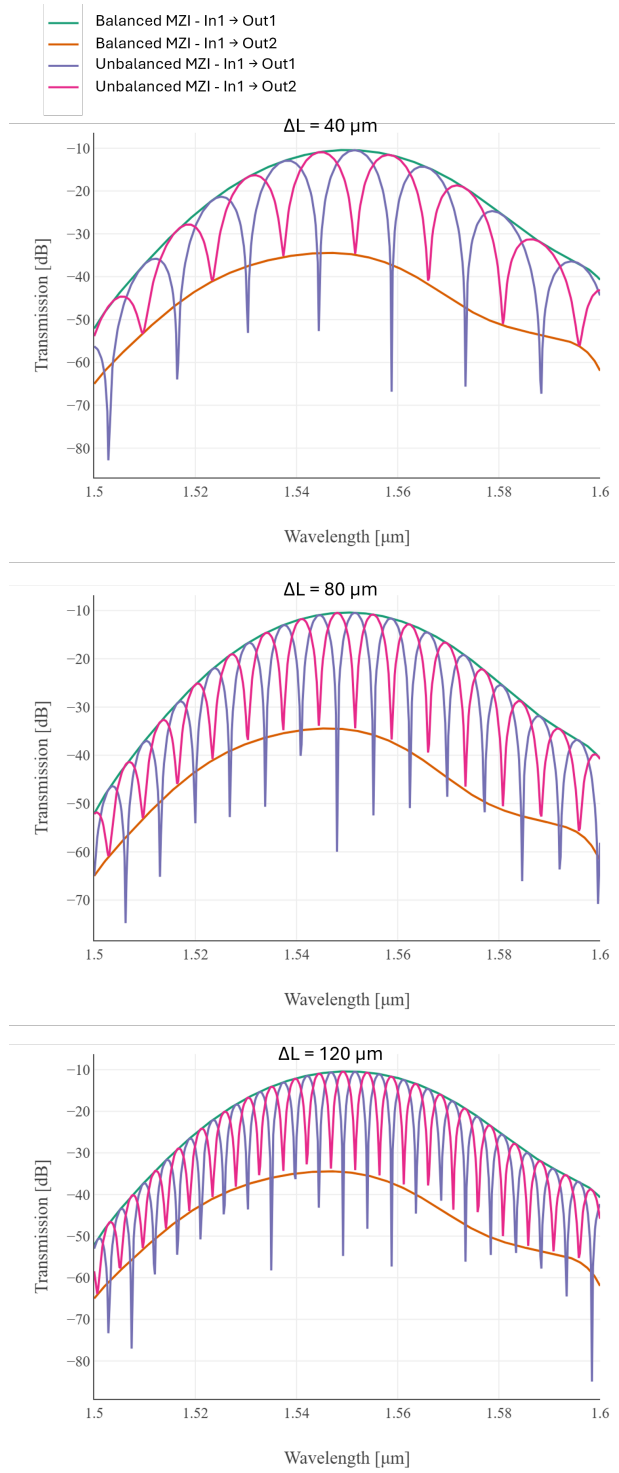

Fig. 4. Simulated transmission spectra of both outputs of the asymmetric MZI for different delay lengths. The simulated spectra of the zero-delay MZI were also plotted for comparison.

B. Mach-Zehnder Interferometers

The proposed design will contain interferometers with three different delays, as well as a fourth interferometer with zero delay that will be used for comparison. Table III-B lists the chosen delay values and the corresponding expected FSRs.

ΔL	FSR
40 µm	14.3 nm
80 µm	7.15 nm
120 µm	4.77 nm

The expected spectral responses for each interferometer were simulated using the simulation tools provided by Luceda Ipkiss. Figure 4 show the results of these simulations for each delay value. In each of the three graphs, we plotted the transmission spectra through both outputs of the corresponding interferometer, as well as the spectra obtained for the zero-delay case. As expected, we see the output power oscillating between maxima and minima, in alternation between the two outputs of the interferometer. The oscillation period grows shorter for longer delays, which is also in line with expectations.

IV. Conclusion

This document outlines a design proposal for asymmetric Mach-Zehnder interferometers of three different lengths. The expected performances of the chosen waveguide geometry and circuit layout were simulated using Lumerical and Luceda Ipkiss. The report includes the expected transmission spectra for both outputs of each device. Once these structures are fabricated and tested, we will compare the measured spectra to the simulated one and estimate the waveguide group index by fitting equation (9) to the measured values of FSR.